Bijux Pollenomics: evidence and open decisions

Nordic pollen, source custody and scientific limits

Bijux Pollenomics

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Source-bound research briefing

What the project can answer today

- Trace pollen, chronology, geography and other source records to governed outputs.
- Reproduce selected published author calculations and locate genuine archive differences.
- Test observed-pollen transfer for Denmark, Finland, Norway and Sweden on a declared taxon axis.
- Run a bounded English LRA reconstruction and forward-pollen prediction study.

These are distinct evidence claims; none is an independent measured-vegetation accuracy result.

Evidence: `reports/README.md`

Four Nordic countries: observed pollen

Target	Sites	Samples	Scored bins	Best fixed baseline, site-macro MAE
Denmark	1	91	24/24	nearest: 0.023242
Finland	28	1,797	435/435	nearest: 0.017335
Norway	52	3,544	1,084/1,178	inverse-distance-squared: 0.019459
Sweden	3	228	51/54	inverse-distance-squared: 0.028740

Country withheld from training; 31 author pollen targets. Previously viewed outcomes make this development evidence. Denmark has one site.

Evidence: `reports/science/denmark-finland-norway-author-pollen.md; reports/science/sweden-author-pollen.md`

English LRA: a real reconstruction, bounded interpretation

- 15 author sites, 1250–1750 BP, 30 pollen targets after a declared transport exclusion.
- 14 sites have three frozen forward-pollen predictions each; one fold refuses after three training optimizations fail.
- No measured vegetation target is available for reconstruction accuracy.

This demonstrates an executable source-bound workflow, not a Nordic LRA validation.

Evidence: `reports/science/english-lra-pollen-prediction.md`

On matched English pollen targets, direct methods score lower

Method	Pollen MAE		Hellinger	
	Direct	LRA forward	Direct	LRA forward
Training mean	.020496	.026273	.282082	.360671
Nearest site	.019552	.021132	.255767	.286001
Inverse distance squared	.018579	.021088	.243673	.285415

14 paired sites, 30 taxa; site-macro development scores. Training information differs between routes. Lower pollen error does not establish better vegetation reconstruction.

Evidence: `reports/science/pollen-prediction.md`;

`docs/report/pollen-reconstruction/author-lra-constrained/direct-pollen/paired-hellinger.json`

Published calculation and archived product are different questions

- Python matches independent author-R replay on 14,074 supported means/SE values; maximum fractional residuals 1.84×10^{-13} and 7.16×10^{-14}.
- The exact 100–350 BP archive window differs from the author replay by up to 9.20 and 10.55 percentage points in the two grids.
- Only one of 25 window intervals matches exactly. Source revisions and PFT transformation remain unresolved.

Numerical correspondence is established for the selected author examples; full historical paper reconstruction is open.

Evidence: `reports/science/publication-correspondence.md`

The next decisions need source evidence

1. Bind Swedish and other Nordic cores to selected ages, historical basin geometry, count denominators and permitted PPE use.
2. Obtain a qualified independent vegetation reference before claiming ecological accuracy.
3. Authenticate historical paper input revisions and the archive PFT/uncertainty transformation.
4. Freeze a new cohort before viewing outcomes for independent confirmation.

Evidence: `reports/needed/README.md`